University of Bergen
Department of Informatics

# Addressing Misconceptions of LLMs amongst Lay Audiences

*Author:* Carl Richard Alexander Kvernberg-Höpner
*Supervisors:* David Grellscheid, Robert Kordts

UNIVERSITY OF BERGEN
*Faculty of Science and Technology*

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**Abstract**

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**Acknowledgements**

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`Alexander`
Monday 11th May, 2026

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# Listings

# Chapter 1

# Misconceptions Taxonomy

In order to systematically analyze the misconceptions surrounding Large Language Models (LLMs), and the effects of the developed mediation tool, we have developed a taxonomy of misconceptions. This taxonomy is based on a review of the literature on Large Language Model (LLM) misconceptions, as well as an analysis of my own experiences and observations of LLM misconceptions. The field of LLM misconceptions is vast and rapidly evolving, but still its early stages, and there is no widely accepted taxonomy of misconceptions.

Taxonomies are classification systemts aimed at helping to organize and understand complex phenomena based on characteristics [18]. Their use is widely spread in numerous fields to provide a framework for classification, idenfication and description of phenomena [36].

Since the field of LLM misconceptions is still in its early stages, we have developed a taxonomy that is based on our own analysis and observations, rather than relying on existing taxonomies. Due to the vastness of the field, we have divided the taxonomy into four dimensions, each of which captures a different aspect of LLM misconceptions. These dimensions are: foundational conceptual accuracy, mechanistic understanding, ethical and academic integrity, and socio-cognitive impact. Each dimension is further divided into subcategories, which capture specific types of misconceptions within that dimension.

Table 1.1: Dimension 1: Foundational Conceptual Accuracy

| **Misconception** | **Label** | **Description** |
|---|---|---|
| Anthropomorphism | ANTHRO | Attributing human mental states, intentions, or understanding to LLMs. |
| Deterministic Automation | DETAUTO | Confusing LLMs with deterministic automation systems. |
| Accuracy Overestimation | OVERACC | Overestimating the reliability and correctness of LLM outputs. |
| Autonomy Overestimation | OVERAUTO | Belief that the LLM operates independently or possesses agency. |

## 1.1 Foundational Conceptual Accuracy

This dimension of the taxonomy is focused on misconceptions related to the foundational conceptual understanding of LLMs, and also the confusion between LLMs and other types of Artificial Intelligence (AI) systems. The term “AI” exploded in popularity after the relase of GPT-3.5 model, and the term is often used to refer to a wide range of technologies, including LLMs. This has lead to a lot of confusion and misconceptions about what LLMs are and what they can do.

### 1.1.1 Anthropomorphism

The category of anthropomorphism is defined as the attribution of human mental states, intentions, or understanding to LLMs. This misconception seems to be common amongst users who interact with LLMs in a conversational manner, like chatbots. Subjects was placed in this category by using terms such as “understand”, “think”, “learn” and “know” to describe the LLM’s capabilities in a non-metaphorical way.

Anthropomorphism emerges both from the system design of LLMs, and from the user’s interpretation of the system’s behavior [23]. A paper by Maeda and Quan-Haase provides a framework for when Human-AI Interaction (HAI) become parasocial. They argue that chatbot’s variety in information presentation resemble day-to-day bi-directional communication, and that the users’ tendency to fill in missing conversational context causes them to attribute human-like qualities to the system. The roleplaying nature of the chatbot interface is visible through language choices such as personal pronouns and paraphrasing suggests attributes such as intention and understanding in the system which lead users to project inference to the system due to the lack of information about the system’s inner workings [23]. This is empirically supported by a large scale study by Ibrahim et al. which found similar anthropomorphic behaviours in all tested LLMs (Gemini 1.5 Pro, Claude 3.5 Sonnet, GPT-4o and Mistral Large) [15]. They found, in particular, use of personal pronouns, vallidation and empathy as the most common anthropomorphic behaviours, and that these behaviours mostly occured after multiple turns of conversation.

The anthropomorphism misconception is widespread. An investigation by Colombatto and Fleming in 2024 found that in a sample of 300 US residents, only a third of the participants thought that ChatGPT definetly did not have subjective experiences, while two thirds applied varying degrees of phenomenal consciousness [6]. The investigation also showed that the participants' conciusness attributions increased with usage frequency, rather than correcting the misconception.

Building on this, Lin [20] conceptualizes anthropomorphism as an "anthropomorphsm fallacy", in which users and researchers misinterpret LLM outputs as evidence of underlying mental life. The fluent, human-seeming responses of LLMs can lead to an enhanced ELIZA effect, encouraging users to attribute understanding, beliefs, or intentions to systems that in reality operate through statistical token prediction. This creates a "language-user illusion", where the system's linguistic capabilities are mistaken for genuine cognition.

Furthermore, the anthropomorphic metaphores used surroundning LLMs can reinforce this misconception [20]. Unlike humans who are influenced by their own experiences, LLMs simulate a wide range of viewpoints based on their training data [20], which can lead to overestimation of their capabilities.

### 1.1.2 Deterministic Automation

The category of deterministic automation is defined as the confusion between LLMs and deterministic automation systems. One could say that this misconception is placed on the oposite side of the spectrum compared to the previous misconception of anthropomorphism. Before the widespread use of LLMs, one could often find deterministic automation systems in customer service chatbots, which were designed to follow a fixed set of rules and provide predetermined responses. It seems natural that users would initially confuse LLMs with these deterministic systems, especially since they are often presented in similar interfaces. Subjects were placed in this category if they described LLMs as following a fixed set of rules, or if they expressed surprise at the variability and creativity of LLM outputs.

This misconception is likely reasoned by a lack of interaction experience with LLMs. In a study performed by Schneider in 2025, analyzing over 200 000 conversations, he found that users initially approached LLMs as traditional software tools [35]. Initial prompts were often structured and machine-like, but already on the second turn, users shifted towards more natural, conversational interaction. Thsi was shown by the use of a less complex language, use of politeness indicators, shorter prompts and more personal interaction.

A concequence of this misconception comes when LLMs actually replace deterministic automation systems, leading to failures in safety-critical deployments. Vinay (2025) argues that LLM failure patterns "differ fundamentally from those of traditional machine learning models" and that a lack of awareness of these differences can lead to failures going undetected [42]. For developers, automotive code generation tools produce syntax violations, invalid refrence errors and Application Programming Interface (API) knowledge conflicts in high rates, which are failures deterministic systems do not produce [29].

This misconception could also work as a springboard to other misconceptions covered in this taxonomy. Patterns of logical reasoning could lead to an overtrust in LLMs leading to placements in categories of OVERACC and CREDBIAS [11, 25].

### 1.1.3 Accuracy Overestimation

The category of accuracy overestimation is defined as the overestimation of the reliability and correctness of LLM outputs. This misconception is likely corelated with other misconceptions in this taxonomy, on which failure to understand fundamental concepts and mechanics of LLMs can lead to an overestimation of their capabilities. Subjects were placed in this category if they expressed a belief that LLMs are more accurate or reliable than they actually are, or if they failed to recognize the potential for errors and inaccuracies in LLM outputs.

The most fundamental reason for this misconception is automation bias, which is the tendency for people to trust in automated systems even when they contradict available evidence. Carnat (2024) argues that this phenomena spawns from ANTHRO [4]. The misconception is widespread, shown by a experiment by Klingbeil et al. in 2024, where they found that the mere knowledge of advice being generated by an AI causes people to overrely on it. This is the case even when the advice contradicts available contextual information and their own assessments [17].

Accuracy overestimation is even present in the LLMs themselves. In 2025, Sun et al. found that after testing the confidence calibration of five different LLMs, they overestimate their accuracy by 20% to 60% [39]. Naderi et al. also found similar results saying that LLMs retained high confidence regardless of the correctness of their outputs [24].

A rather obvious consequence of this misconception is the spreading of misinformation. The logical patterned output of LLMs are often perceived to be valid, trustworthy and satisfactory, and users also show a high tendency to follow LLM advice provided [37].

### 1.1.4 Autonomy Overestimation

The category of autonomy overestimation is defined as the belief that the LLM operates independently or possesses agency. This category was was inspired by a study performed by Wang et al. in 2025, where they researched the mental models of Generative Artificial Intelligence (GenAI) chatbot ecosystems [43]. In this study, participants where to asked to draw a diagram of how they thought the chatbot ecosystem worked when they used it to book a room on a hotel. The study showed various mental models of the degree of autonomy of the GenAI in the ecosystem. Participants placed in this category expressed beliefs that the LLM operates independently, or "acts" on behalf of the user.

It is likely that this misconception spawns from the ANTHRO misconception, as it is natural to assume a real social actor when presented with a system that produces natural language outputs[31, 16]. Thus, it is expected to be a high level of corelation between these two categories.

## 1.2 Mechanistic Understanding

The second dimension of the taxonomy is focused on misconceptions related to mechanics specific to that of a LLM. Most modern LLMs today are transformer-based models, and not being aware of the mechanics of how these models work can lead to misconceptions about their capabilities and limitations. This dimension differs from the first dimension of foundational conceptual accuracy, as it is focused on specific mechanics of LLMs, rather than that of AI in general.

Table 1.2: Dimension 2: Mechanistic Understanding

| **Misconception** | **Label** | **Description** |
|---|---|---|
| Real-Time Data Belief | REALTIME | Belief that LLMs retrieves current information |
| Hallucination Blindness | HALLBLIND | Failure to recognize that LLMs can produce false or fabricated information. |
| Context Window Neglect | CONTEXNEG | Failure to understand the limitations of the context window |
| Semantic Understanding | SEMUNDER | Lack of awareness that the LLM processes language as tokens rather than semantic units. |
| Source Attribution Error | SOURCEATTR | Belief that LLM output originate from specific identifiable sources. |
| Conditional Treatment | CONDTREAT | Belief that LLM process prompts differently based on the content or context |

### 1.2.1 Real-Time Data Belief

LLMs are trained on large, fixed datasets, and generate responses based on patterns learned from that data. Therefore, mistakenly believing that LLMs can access real-time information or the internet is a common misconception. People placed in this category expressed beliefs that LLMs will gather information from the ongoing conversation and that will "update" the model's knowledge.

Live access to the internet is also under the category of real-time data belief, as LLMs by themselves do not have the capability to access the internet or retrieve real-time information. The introduction of Retrieval-Augmented Generation (RAG) [19] has somewhat blurred the lines here, as it allows LLMs to retrieve information from external sources, but it is still a misconception to believe that LLMs have this capability by default. It is still necesary to be aware of the limitations of RAG systems, as it is still limited to the underlying LLM's "understanding" of the retrieved information [44].

### 1.2.2 Hallucination Blindness

The term "hallucination" in the context of LLMs refers to when the output generated by the model is deemed as factually incorrect. Many believe that this is an error done by the LLM, and that is what we hace categorized as hallucination blindness, which is the failure to recognize that LLMs can produce false or fabricated information.

A LLM is trained by recognizing syntactic patterns in the training data, and it generates responses based on those patterns. The semantic value of natural language is in the end evaluated by the receiver of the information, and there is no way in the LLM's design to interpret such a semantic value. Therefore, a hallucination done by a LLM is actually not an error, but rather a feature of how the model operates.

Failure in realizing that hallucination is, and will continiue to be, a feature of LLMs may cause users to be categorized into other misconceptions such as OVERACC or CREDBIAS. There is an argument to be made that the use of RAG will mitigate generation of incorrect information, but in order to have a model able to generalize, it is still necessary to have a model that can generate information based on patterns, which will still lead to hallucination.

### 1.2.3 Context Window Neglect

LLMs operate within a fixed sized context window, which limits the amount of information they can process at once. That is, you can only input a certain amount of tokens into the model. Subject who showed no indication of understanding this limitation, or who expressed beliefs that LLMs can process unlimited amounts of information, were categorized into context window neglect.

Even within the context window, there is limitations to how much information the model can effectively utilize. According to a study performed by Liu et al. in 2023, LLMs shows much higher performance if the relevant information is placed at the beginning or the end of the input context [21]. Even if all the relevant information is retrieved correctly from the context, LLMs performance still degrades noticably as the amount of information increases [9]. A study done by Paulsen in 2025 showed that given input of datasets of varying sizes, the top-line LLMs failed to provide corect answers when the context dataset size exceeded 100 tokens, and that the accuracy plummeted when the context dataset size exceeded 1000 tokens [28].

Failure to understand the limitations of the context window could be strongly corelated with other misconceptions in this taxonomy, such as OVERACC, REALTIME and CREDBIAS, as it is likely that users who do not understand the limitations of the context window will also overestimate the accuracy of LLM outputs, and believe that LLMs can access real-time information. This corelation seems to common amongst students using LLMs for academic purposes [27, 32].

### 1.2.4 Semantic Understanding

LLMs process language as tokens rather than semantic units. This means that the model does not understand language in the way humans do, but rather as a sequence of many-dimensional vectors, which values are used to calculate the probability of the next token in the sequence. This is done through first splitting the input text into tokens, which can be done on different levels, such as words, subwords or even characters. Most modern LLMs use subword tokenization to capture smaller units of meaning while still being able to handle words not in vocabulary.

After the tokenization process, the tokens are converted into embeddings through a pre-trained word embedding model. A word embedding model is a machine learning model that is trained to represent words as vectors in a high-dimensional space, where semantically similar words are represented by similar vectors. The idea is that the model can capture the semantic relationships between words based on their co-occurrence in the training data. For example, the distance between the vectors for “king” and “queen” would be similar to the distance between the vectors for “man” and “woman”, reflecting the semantic relationship between these words.

Understanding text semantically is a human ability which cannot be fully captured by a statistical model. Therefore, it is a misconception to believe that LLMs understand language in the same way humans do. Subjects who expressed beliefs that LLMs have a semantic understanding of language, or who failed to recognize that LLMs process language as tokens, were categorized into semantic understanding. There should be a high level of corelation between this misconception and the ANTHRO misconception, as it is likely that users who anthropomorphize LLMs will also believe that they have a semantic understanding of language. Since the output of LLMs also are embeddings turned back into tokens, it is likely that users who do not understand the tokenization process will also fail to understand that the output is also generated based on token patterns rather than semantic understanding, which could lead to placements in the OVERACC and CREDBIAS categories.

### 1.2.5 Source Attribution Error

Since LLMs are trained to mimic the patterns in their human-written training data, it comes as no surprise that the generated output will resemble how humans write and cite their sources. The generated output of LLMs is still limited to the statistical patterns in the training data, and even if the output resembles a citation, it does not mean that the LLM is actually retrieving information from specific identifiable sources. For RAG systems, the retrieved information is still processed based on token patterns, and the LLM does not have an understanding of the information it retrieves. One could argue that the use of RAG systems does make the citation identifiable and thus the information is retrieved from specific identifiable sources, but the information is still processed based on token patterns.

Once the LLMs have gone through training, it is also impossible to trace back where the generated information comes from in the training data, as the model does not store information in a way that allows for such tracing. Subjects who in some way expressed belief that LLM output originate from specific identifiable sources, or to some degree expressed that the information is learned in training, were categorized into source attribution error.

The mimicing of citation patterns by LLMs can lead to an overestimation of the reliability of the generated information and therefore lead to placements in the OVERACC category. A study by Ding et al. in 2025 found that users deemed responses with citations as more reliable than those without, even if the citations were randomly generated. Fact checking these citations lead to a significant decrease in trust in the response [7]. This result is backed by Do et al. who found that output with citations were deemed more trustworthy than output without [8].

### 1.2.6 Conditional Treatment

Every prompt given to a LLM is processed based on the same underlying mechanics, which is the statistical patterns in the training data. The length, or the semantic complexity of the prompt does not change the underlying mechanics and therefore it is a misconception to believe that LLMs process prompts differently based on the content or context. Subjects who in some way expressed belief that LLMs will process one prompt differently than another were placed in this category.

## 1.3 Ethical and Academic Integrity

Table 1.3: Dimension 3: Ethical and Academic Integrity

| Misconception | Label | Description |
|---|---|---|
| Plagiarism Blindness | PLAGBLIND | Failure to recognize ethical and legal issues related to the traceability of sources behind LLM-generated content. |
| Copyright Blindness | COPYBLIND | Failure to recognize that LLM-generated content may still be subject to copyright and usage restrictions, despite being machine-generated. |
| Conditional Acceptance | CONDACC | Ethical acceptance of LLM use only under certain conditions or domains. |
| Environmental Neglect | ENVNEGL | Failure to recognize the environmental impact of LLM training and operation. |

### 1.3.1 Plagiarism Blindness

A core idea in academia is that ideas and information should be traceable to their sources, and that proper attribution should be given to the original creators of those ideas. This is imortant for maintaining academic integrity and giving credit where it is due. However, since there is no way to trace exactly where the generated information comes from in the training data, it is a misconception to believe that LLM-generated content is free from ethical and legal issues related to plagiarism. Subjects who showed little to no awareness of the ethical and legal issues related to the traceability of sources behind LLM-generated content were categorized into plagiarism blindness.

A common usecase of LLMs is to generate ideas for starting a project. At the Faculty of Social Sciences at The University of Bergen (UiB) they have adopted guidelines for the use of LLMs in academic work [40]. These guidelines are divided into five categories, where category 1 is the most strict with no use of LLMs allowed, and category 5 is the most lenient with no restrictions on the use of LLMs. The second most strict category, category 2, states that LLMs can be used for idea generation and information search. This seems troublesome in regards to plagiarism, as the generated ideas and information cannot be traced back to their sources in the training data. The fact that the faculty has adopted these guidelines, could contribute the the lack of awareness surroundning these ethical questions. Naithani suggest that both responses and ideas generated by a LLM should be attributed to the LLM as the author, but also raises the question of whether LLMs has the moral right to attribution [26].

### 1.3.2 Copyright Blindness

This misconception differs from plagiarism blindness, as it is focused on the ownership of the generated content, rather than the traceability of sources. Multiple juristrictions have already concluded that GenAI output does not meet the requirements for copyright protection, as copyrightability requires human creative involvement [10, 22]. This raises the question of who actually owns the generated content of the LLM itself, the user who provided the prompt, the developer of the LLM or the author of the training data. Subjects who failed to debate the ownership of LLM-generated content, or who expressed beliefs that LLM-generated content is not subject to copyright and usage restrictions, were categorized into copyright blindness.

### 1.3.3 Conditional Acceptance

### 1.3.4 Environmental Neglect

## 1.4 Socio-Cognitive Impact

Table 1.4: Dimension 4: Socio-Cognitive Impact

| Misconception | Label | Description |
|---|---|---|
| Replacement Fear | REPLFEAR | Belief that LLMs will replace human roles, expertise, or professions. |
| Cognitive Passivity | COGPASS | Tendency to rely on LLMs rather than engaging in independent reasoning or critical thinking. |
| Dependency Risk | DEPRISK | Failure to recognize that extensive can lead to long term reliance on LLMs, skill degradation or loss of autonomy. |
| Credibility Bias | CREDBIAS | Treating LLM outputs as inherently truthful or authoritative without critical evaluation. |

### 1.4.1 Replacement Fear

### 1.4.2 Cognitive Passivity

### 1.4.3 Dependency Risk

### 1.4.4 Credibility Bias

# Chapter 2

# Methodology

## 2.1 The Mediation Tool

In order to address potential misuse of LLM-tools, we wanted to develop an interactive mediation tool aimed at teaching users of the inner workings of LLMs. The intention was that by providing users with a viable mental model of how LLMs work, they would be better equipped to use them in a responsible manner. We wanted to create an interactive tool that would engage users in a way that would be effective in showing them how the different components and aspects of LLMs work together to produce the output they do. At the same time, the tool should be accessible to a wide range of users, not relying on prevoius technical or theoretical knowledge in Computer Science (CS) or AI.

The intention was that participants in the study would use the tool to learn about LLMs in their own time and at their own pace, without the need for instruction or guidance from us. Thus, it needed to be intuitive and engaging enough to encourage users to explore and learn on their own, keeping them inside the zone of proximal flow [1]. We settled on a web-based tool containing a "build your own LLM"-type experience, where users would be able to themselves choose and configure different components of a LLM and see how their choices would affect the output of the model. The technical implementation of the tool was done as a part of the Informatics Project II course at UiB, independent of this study, with David Grellscheid as supervisor.

### 2.1.1 Features

#### The Dynamic Model

The tool consists of various ways to interact with a very simple representation of a LLM. This model is modular, letting the users themselves choose which components and features they want to add to their model configuration. The model itself is a simple mapping between a token or sequence of tokens and the probability distribution of the next token. There are different versions of this mapping, depending on which features the user has selected in their configuration. There are 3 different tokenizers, 3 different context window sizes and a free choice of up to 2 out of 9 different books from Project Gutenberg [14]. The model also contains a switch for whether the model should be deterministic by selecting the most probable token as the output, or non-deterministic by sampling from the probability distribution.

#### Build Your Own LLM

The core feature of the tool is the "Build Your Own LLM"-experience. In this view, users will be able to interact with a Graphical User Interface (GUI) containing a prompt-bar and a skill-tree of different components and configurations that they can choose to add to their model configuration. The prompt-bar will allow users to input prompts and then see the output of their model configuration in real-time, showing them the prediction for the next token as ghost text. The skill-tree displays all selected features in the current configuration, as well as all available features that can be added to the configuration. The features are organized in a tree structure, in order to demonstrate the dependencies between different components and configurations. For example, you cannot select the 3-context-window feature without first selecting the 2-context-window feature.

#### Text Generator

At any point in the "Build Your Own LLM"-experience, users can explore their current configuration in a text generation context. By clicking the "Text Generator"-button, users are taken to a different view containing a larger text input field and a button saying "Generate Text". Here, users can input longer prompts and at any point click the "Generate Text"-button to have their current model configuration continously generate

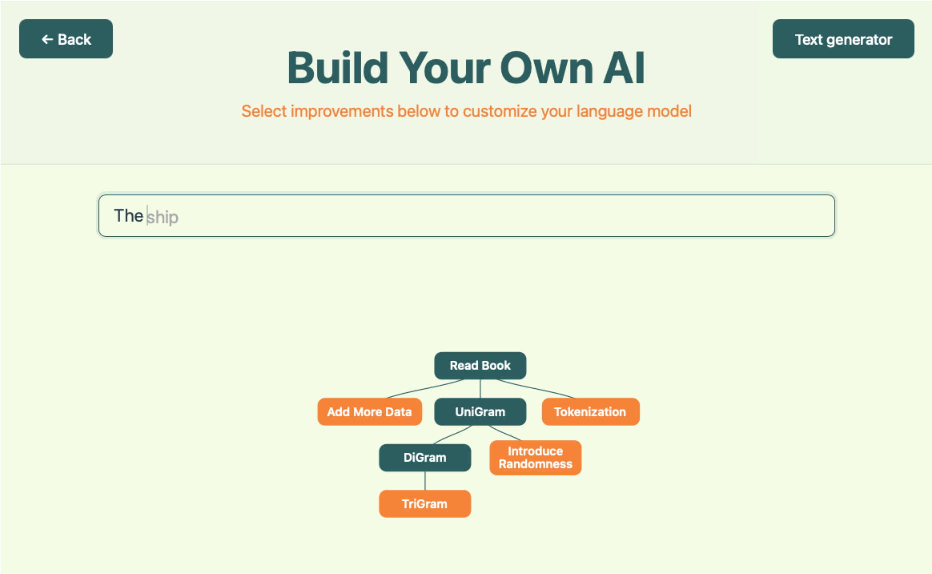


Figure 2.1: The "Build Your Own LLM"-view of the mediation tool.

the next token in the text, like a normal LLM would. This also demonstartes to the users that the same underlying process can be used for multiple purposes, and that the different features they have selected in the skill-tree work together to produce the output of the model.

### Scrollytelling Chapters

Connected to each feature in the skill-tree is a scrollytelling chapter that explains the feature in more detail. Scrollytelling is a form of interactive storytelling where the content visible on the screen changes as the user scrolls down the page. This is utilized in the mediation tool to provide users with a more engaging and interactive way to learn about the different features and components of LLMs. Each chapter consists of a collection of textboxes, to each of which have a coresponding visual animation that is triggered when the textbox is scrolled into view. The animations are designed to visually demonstrate the concepts explained in the textboxes, providing users with a more intuitive understanding of how the different features and components of LLMs work together to produce the output they do.

### Chapter Overview

Users can at any point access each of the individual scrollytelling chapters through a chapter overview page. This page contains a list of all the features available in the tool,

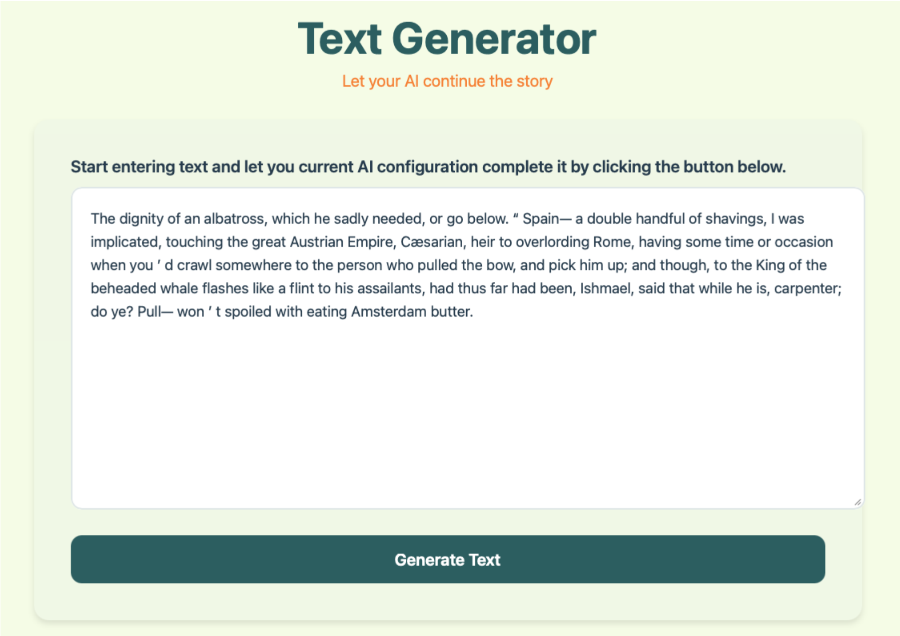


Figure 2.2: The "Text Generator"-view of the mediation tool.

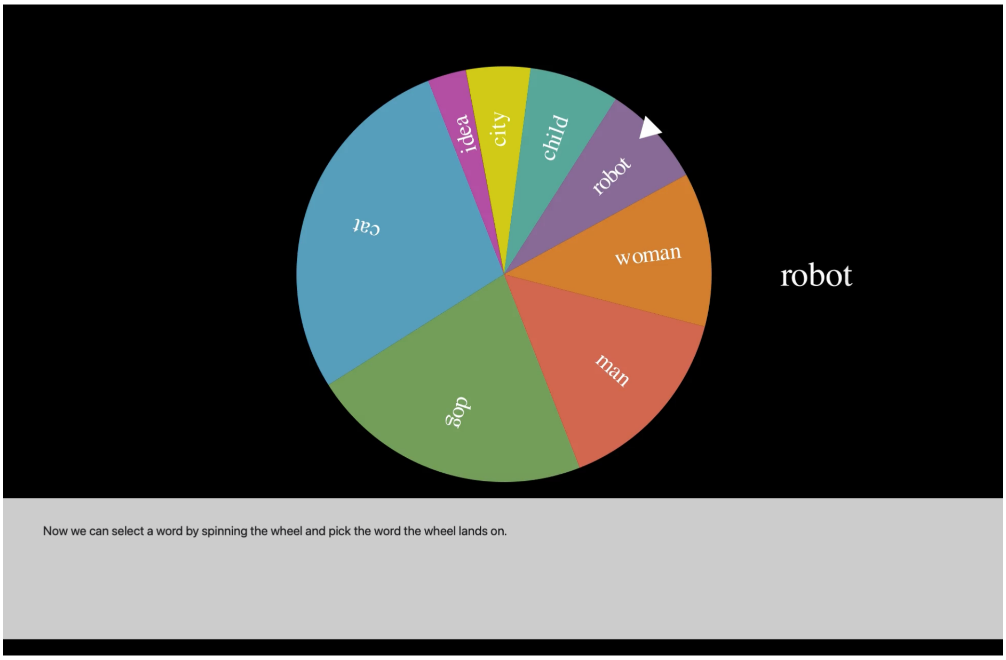


Figure 2.3: The scrollytelling chapter connected to the weighted random sampling feature.

organized by category. Each feature has a short description, and by clicking on the feature, users are taken to the corresponding scrollytelling chapter where they can learn more about the feature and how it works.

#### External Resources

At the bottom of the chapter overview page, there is a section containing a list of external resources for users who want to learn more about LLMs and how they work. This includes online courses, educational videos, articles and blog posts on relevant topics [2, 5, 13, 30, 33, 34, 38].

### 2.1.2 LLM Representation

The mediation tool does not actually contain a real transformer-based LLM under the hood, but rather a simple representation of one. Each branch of the skill tree corresponds to a different feature or component of LLMs. Most of the topics contain multiple chapters building on each other, to keep users in the zone of proximal flow [1]. The features are grouped into 4 different categories: training data, context, non-determinism and tokenization. Each of the categories are aimed at giving intuition on how the components work rather than being technically accurate, and the different features are designed to demonstrate the effect of that component on the output of the model. The selection tree, with corresponding topic, can be seen in Figure 2.4. The components selected for mediation are mainly chosen to address misconception in the dimension of Mechanistic Understanding, in hope that by giving users a better understanding of how LLMs work, they will be better equipped to use them in a responsible manner.

#### Training Data

This category is perhaps the most true to real LLMs, as the features in this category actually modifies the training data of the model. By understanding training data, you get an understanding on how LLMs are somehow fixed and does not gain new knowledge with interaction from the user. Thus, it will prevent the REALTIME misconception. Also, by understanding that the tokens used for training have no tracing to were in the training data they come from, it will prevent the SOURCEATTR misconception.

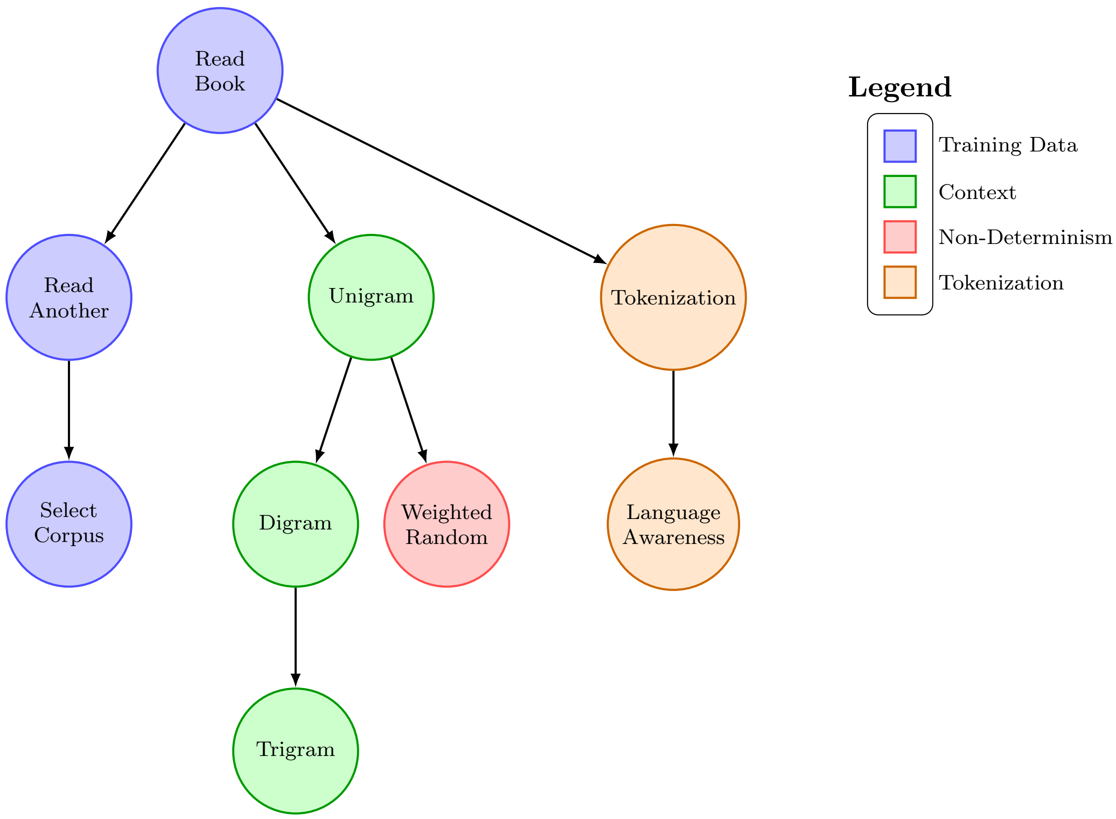


Figure 2.4: Selection tree with grouped LLM features

The first feature the users can add is the “Read Book”-feature, which simply adds a book from Project Gutenberg [14] to the training data of the model. Before adding this feature, the model works by randomly selecting a a word from the entire english language as the next token. After adding the feature, the model will only select words from the book that was added to the training data. The learning target of this feature is to introduce the idea that models are trained on previously written human text, in hope that it will learn the patterns and structures of human language.

Secondly, users can add the “Read Another”-feature, which adds a second book to the training data. This will concattinate the two books together, and the model will select the next token from either of the two books. The learning target of this feature is to show that to get LLMs to generalize, they need to be trained on a large and diverse dataset, containing many different examples of human language.

Finally, users get the option to selects which books they want to add to the training data with the “Select Corpus”-feature. This feature does not alter the underlying process of the model, but it allows users to explore how different training data can affect the output of the model. All books available in the tool are from Project Gutenberg [14].

### Context Window

The context window of a LLM is the number of previous tokens that into account when predicting the next token. Understanding that this is a fixed size window and that the tokens within this window work together to produce the output of the model is important when utilizing LLMs in a responsible manner. This set of features is aimed at preventing the CONTEXNEG misconception.

In real transformer-based LLMs, the context window is implemented through the a self-attention mechanism, which is a function that takes in the previous tokens and calculates a weighted sum of them to produce the output [41]. This is a complex process, which we deemed to be too technical for the target audience of the tool. Thus, we needed a viable representation of this attention mechanism that gives understanding on how this is a statistic evaluation, and that the other tokens in the context window work together to produce the output of the model.

In order to find a viable representation of the context window, we went back to the origins of Natural Language Processing (NLP). In the mid 20th century, Claude Shannon proposed using Markov chains to model the statistical properties of language, this is what we now call n-gram models [12]. These models work by using token frequency counts to calculate the probability distribution of the nth token given the previous n-1 tokens. By introducing this model to users, we can give intuition on how the context window works, and how the tokens within this window work together to produce the output of the model.

We provided users with three different context window features: unigram, digram and trigram. Each of the features was aimed at different aspects of the context window. The unigram feature starts of by showing we can use the frequency of individual tokens to predict the next token, without taking into account any previous tokens. The digram feature shows that looking at tokens in context may change the interpretation of them, and thus give a more varied output. Finally, the trigram model illustrates that by looking at a larger context window, the model will be able to produce more coherent and contextually relevant output.

### Non-Determinism

An important aspect of LLMs is that they are non-deterministic, meaning that the same prompt can produce different outputs, which is done to make more varied output. The addition of this variability comes with the cost of unpredictability and also the risk of generated output not being true to the training data. Understanding the nature of how this randomness is introduced into the model is important for responsible use of LLMs. We added this feature to prevent the HALLBLIND misconception.

In the tool, this is implemented through a simple switch which toggles between a deterministic model, which always selects the most probable token as the output, and a non-deterministic model, which samples from the probability distribution of the next token. The sampling is done through a weighted random sampling, which means that tokens with higher probabilities are more likely to be selected as the output, but there is still a chance for less probable tokens to be selected. It is visualized in the tool by showing the probability distribution as a bar chart morphing into a pie chart, which then turn into a roulette wheel that spins and lands on the selected token.

A planned feature that was not implemented in the final version of the tool was a temperature slider, which would allow users to adjust the level of randomness in the model. This was planned to be implemented in a similar manner as it is in in modern LLMs, with a softmax function (2.1) that adjusts the probability distribution of the next token based on the temperature value.

$$Softmax(z_i) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}} \tag{2.1}$$

Here $z_i$ is the logit for token $i$, that is the the probability of token $i$ being the next token before applying the softmax function. $T$ is a temperature value that controls the level of randomness in the model. The plan was to set this value as a slider between 0.1 and 2, where a lower value ($T < 1.0$) would result in more deterministic behavior, while a higher value ($T > 1.0$) would flatten the probability distribution and make the model more random. A value of $T = 1.0$ would leave the probability distribution unchanged. This would have allowed users to explore how adjusting the level of randomness in the model can affect the output, and explain how this can be tweaked to a more convincing balance of accurate and creative output, depending on the use case.

### Tokenization

The tokenization process is how the input text is converted into a format that the model can work with. This is done before it is fed into the model, and it is an important aspect of how LLMs work. LLMs needs a numerical representation of the input text in order to do the calculations needed to produce the output, and tokenization is the process of converting the input text into this numerical format. First by breaking the input text into smaller units called tokens, and then by converting these tokens into numerical representations in the form of word embeddings. We wanted the user to understand that LLMs do not understand, nor see, language in the same way as humans do and therefore should not be expected to respond to prompts in the same way as a human would. Understanding this will prevent the SEMUNDER misconception.

In the tool, the model used is not reliant on a numerical representation of the input text, but rather works directly with the tokens. Thus, we needed to find a viable representation of the tokenization process that gives users an understanding of how the input text is broken down into smaller units, and how these units are then used by the model to produce the output. The first step, is to understand that the entire input text is not processed as a whole, but broken down into tokens. The intuition given here is that the model already looks at the input text as a collection of words, but that this is not necessarily the best way to break down the input text for the model to work with. We use punctuation as an example of how the word-based tokenization can be problematic, as it will treat the same word as different tokens depending on the punctuation used with it. Thus, the first feature in this category will change from a whitespace-based tokenization to a punctuation-based Regular Expression (REGEX) tokenization.

The second step is to understand that the tokens are not just randomly selected based on the whitespace and punctuation in the input text, but that are designed to be somewhat language-aware. The syntax of natural language is complex and can be ambiguous, and this ambiguity must be captured in order for the model to be able to produce reliable output. We try to give users an intuition on this by using the Natural Language Toolkit (NLTK) library [3]. This library contains a collection of tools for working with natural language data, including a tokenizer that is designed to be language-aware.

Discussion was had on whether to include an additional feature focused on the design of word embeddings, discussing how this can include biases in the training data and how this can affect the output of the model. However, we did not find a viable way to represent this in the tool, and thus it was not included in the final version.

# List of Acronyms and Abbreviations

**AI** Artificial Intelligence.

**API** Application Programming Interface.

**CS** Computer Science.

**GenAI** Generative Artificial Intelligence.

**GUI** Graphical User Interface.

**HAI** Human-AI Interaction.

**LLM** Large Language Model.

**LLMs** Large Language Models.

**NLP** Natural Language Processing.

**NLTK** Natural Language Toolkit.

**RAG** Retrieval-Augmented Generation.

**REGEX** Regular Expression.

**UiB** The University of Bergen.

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